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Question	Answer	Marks	Guidance
4(a)	$G_X(t) = \frac{2}{5}t + \frac{3}{5}t^2$	B1	CAO
		1	
4(b)	$G_Y(t) = \frac{1}{625}t^4(2+3t)^4 \left[= t^4 \left(\frac{2}{5} + \frac{3}{5}t \right)^4 \right]$	M1	Attempt at 4th power of <i>their</i> $G_X(t)$.
		A1	
		2	
4(c)	Attempt at coefficient of t^6 in <i>their</i> $G_Y(t)$.	M1	Identify required term, coefficient must a non-zero probability. May be seen in part 4(b) .
	$P(Y=6) = \frac{216}{625}$	A1	AWRT 0.346.
		2	

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4(d)	$G_Y'(t) = \frac{4}{625}t^3(2+3t)^4 + \frac{12}{625}t^4(2+3t)^3$ OR $G_Y'(t) = \frac{1}{625}(64t^3 + 480t^4 + 1296t^5 + 1512t^6 + 648t^7)$	M1	Attempt at differentiating <i>their</i> $G_Y(t)$ from part 4(b).
	$G_Y''(t) = \frac{12}{625}t^2(2+3t)^4 + \frac{96}{625}t^3(2+3t)^3 + \frac{108}{625}t^4(2+3t)^2$ OR $G_Y''(t) = \frac{1}{625}(192t^2 + 1920t^3 + 6480t^4 + 9072t^5 + 4536t^6)$	M1	Attempt at differentiating <i>their</i> $G_Y'(t)$.
	$G_Y'(1) = \frac{32}{5} = 6.4, G_Y''(1) = \frac{888}{25} = 35.52$	B1	WWW
	$\text{Var}(Y) = \frac{888}{25} + \frac{32}{5} - \left(\frac{32}{5}\right)^2$	M1	Use correct formula with <i>their</i> values.
	$\text{Var}(Y) = \frac{24}{25} \quad [=0.96]$	A1	CWO
	Alternative method for question 4(d)		
	$E(X) = \frac{2}{5} + \frac{3}{5} \times 2 = \frac{8}{5}$ or use $G_X'(t) = \frac{2}{5} + \frac{6}{5}t$.	M1	
	$E(X^2) = \frac{2}{5} + \frac{3}{5} \times 4 = \frac{14}{5}$ or use $G_X''(t) = \frac{6}{5}$	M1	
	$\text{Var}(X) = \frac{14}{5} - \left(\frac{8}{5}\right)^2$ or $\text{Var}(X) = \frac{6}{5} + \frac{8}{5} - \left(\frac{8}{5}\right)^2$	M1	
	$\text{Var}(X) = \frac{6}{25}$	A1	
	$\text{Var}(Y) = 4\text{Var}(X) = \frac{24}{25}$	A1	CWO
		5	